

Lab - Cybersecurity Case Studies

Objectives

Research and analyze cyber security incidents.

Part 1: Conduct search of high profile cyberattacks.

Part 2: Write an analysis of a cyberattack.

Background / Scenario

Governments, businesses, and individual users are increasingly the targets of cyberattacks and experts predict that these attacks are likely to increase in the future. Cybersecurity education is a top international priority as high-profile cyber-security related incidents raise the fear that attacks could threaten the global economy. The Center for Strategic and International Studies estimates that the cost of cybercrime to the global economy is more than \$600 billion annually. In this lab, you will study four high profile cyberattacks and be prepared to discuss the who, what, why and how of each attack.

Required Resources

- PC or mobile device with internet access

Instructions

Part 1: Conduct search of high profile cyberattacks.

- a. Using your favorite search engine conduct a search for each of the cyberattacks listed below. Your search will likely turn up multiple results ranging from news articles to technical articles.

- The Stuxnet Virus
- Marriott data breach
- United Nations data breach
- Microsoft customer support database breach
- Lifelabs data breach

Note: You can use the web browser in virtual machine installed in a previous lab to research the hack. By using the virtual machine, you may prevent malware from being installed on your computer.

- b. Read the articles found from your search in Step 1a and be prepared to discuss and share your research on the who, what, when, where, and why of each attack.

Part 2: Write an analysis of a cyberattack.

Select one of the high-profile cyberattacks from Step 1a and write an analysis of the attack that includes answers to the questions below.

Questions:

- a. Who were the victims of the attacks?

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b. What technologies and tools were used in the attack?

Sâu máy tính Stuxnet, khai thác 4 lỗi Zero-day trên Windows, chứng chỉ số bị đánh cắp và mã độc can thiệp vào PLC Siemens Step 7

c. When did the attack happen within the network?

Xâm nhập qua USB từ khoảng 2007–2009 (vượt qua mạng air-gap), nằm ẩn theo dõi nhiều tháng trước khi phá hoại và bị phát hiện năm 2010

d. What systems were targeted?

Máy tính Windows nối với hệ thống SCADA/ICS, phần mềm Siemens Step 7 và các máy ly tâm hạt nhân

e. What was the motivation of the attackers in this case? What did they hope to achieve?

Phá hoại tình báo/quân sự nhằm làm chậm chương trình hạt nhân của Iran mà không cần tấn công vũ lực.

f. What was the outcome of the attack? (stolen data, ransom, system damage, etc.)

Phá hỏng vật lý khoảng 1.000 máy ly tâm (ép quay quá tốc độ nhưng báo chỉ số giả về màn hình), làm trì hoãn nghiêm trọng chương trình hạt nhân của Iran